DATA-DRIVEN STRATEGY TO REDUCE HOSPITAL READMISSIONS & OPTIMIZE COSTS

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During this project, I analyzed a real-world hospital dataset (2022–2025, 984 patient encounters across Indian states) to uncover the factors that drive **readmissions**, **rising costs**, and **patient dissatisfaction**.
Using **SQL, Python, Machine Learning, and Power BI**, I translated complex medical data into actionable insights that hospitals can use to improve both **patient outcomes** and **operational efficiency**.

Project Goal

Readmissions are one of the biggest challenges in healthcare. They increase costs, lower satisfaction, and strain medical resources.
My goal was to understand:

- Why patients are being readmitted
- How much financial impact it creates
- Whether patient satisfaction can predict it
- And if a model could identify high-risk patients before discharge

DATA EXPLORATION WITH SQL

Data Table:

```sql

create table Hospital_data (
    Patient_ID INT PRIMARY KEY,
    Age INT,
    Gender VARCHAR(10) NOT NULL,
    Diagnosis VARCHAR(100),
    Medication VARCHAR(100),
    Admission_Date DATE NOT NULL,
    Discharge_Date DATE,
    Patient_State VARCHAR(50),
    Year_of_Admission INT,
    Length_of_Stay INT,
    Readmission VARCHAR(3),
    Outcome VARCHAR(20),
    Satisfaction INT,
    Insurance_Claimed VARCHAR(3),
    Total_Cost DECIMAL(10, 2)
);
```

Table Content:

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Patient_ID	Age	Gender	Diagnosis	Medication	Admission_Date	Discharge_Date	Patient_State	Year_of_Admission
1	45	Female	Heart Disease	Angioplasty	2024-02-07	2024-02-12	Kerala	2024
2	60	Male	Diabetes	Insulin Therapy	2023-03-11	2023-03-14	Goa	2023
3	32	Female	Fractured Arm	X-Ray and Splint	2025-01-02	2025-01-03	Maharastra	2025
4	75	Male	Stroke	CT Scan and Medication	2023-12-30	2024-01-06	Manipur	2023
5	50	Female	Cancer	Surgery and Chemotherapy	2022-05-10	2022-05-20	Mizoram	2022
6	68	Male	Hypertension	Medication and Counseling	2025-06-27	2025-06-29	Sikkim	2025
7	55	Female	Appendicitis	Appendectomy	2023-12-13	2023-12-17	Karnataka	2023
8	40	Male	Fractured Leg	Cast and Physical Therapy	2022-07-28	2022-08-03	Gujarat	2022
9	70	Female	Heart Attack	Cardiac Catheterization	2023-05-11	2023-05-19	Kerala	2023
10	25	Male	Allergic Reaction	Epinephrine Injection	2023-04-25	2023-04-26	Kerala	2023
11	48	Female	Respiratory Inf...	Antibiotics and Rest	2024-03-28	2024-03-30	Delhi	2024

Queries:

```sql
----- Financial Analysis -----
select year_of_admission, round(sum(Total_cost),2) as total_revenue, round(avg(total_cost),2) as avg_cost
from hospital_data
group by 1
order by 1 desc;

select Insurance_Claimed, count(*) AS patient_count
from hospital_data
group by 1;                                          --(Yes-351, No-633)--


----- Readmission & Outcome Metrics -----
Select
    round(sum(case when readmission = 'Yes' then 1 else 0 end) * 100.0 / count(*),2) as readmission_rate
from hospital_data;                                  --(26.83 %)--
```

Diagnosis with Most Patient:

```sql
----- Disease and Treatment Insights -----
select diagnosis, count(*) as total_case
from hospital_data
group by 1
order by 1 desc;                                     --most common disease (Stroke)--

select diagnosis, round(avg(total_cost),2) as Avg_cost
from hospital_data
group by 1
order by Avg_cost desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

diagnosis	total_case
Stroke	66
Respiratory Infection	65
Prostate Cancer	65
Osteoarthritis	64
Kidney Stones	65
Hypertension	66
Heart Disease	65

Before any modeling, I began by exploring and understanding the data structure using SQL.

I wrote queries to:

- Count **total patients** and split them by gender, age group, and region
- Calculate **average stay duration**, **average treatment cost**, and **readmission percentage**
- Identify the most **common diagnoses** and their readmission trends
- Join multiple tables (patients, admissions, outcomes) to validate data consistency
- Find missing or inconsistent values in cost and satisfaction fields

*"**This SQL step gave me a clean and reliable foundation for deeper analysis**."*

STATISTICAL ANALYSIS & VISUALIZATION IN PYTHON

Next, I moved to Python for **Exploratory Data Analysis** (EDA).
Using libraries like Pandas, NumPy, and Matplotlib, I summarized the entire dataset to discover patterns:

Found that **26.8%** of patients were readmitted — roughly 1 in 4.
Compared cost distributions using t-tests:

- **Readmitted:** ₹13,650 (mean)
- Non-readmitted: ₹6,430 (mean)
- The difference was statistically significant ($p < 0.001$).
  - Evaluated satisfaction levels:
- Readmitted: 3.1/5
- Non-readmitted: 3.8/5
- Again, significant difference ($p < 0.001$).
  - Segmented data by age, state, and diagnosis, uncovering that:
- Older patients (50+) and those from states like Punjab and Kerala showed higher readmission rates.
- **Heart-related conditions** (Heart Disease, Heart Attack) were the top contributors.

Basic Libraries:

Hospital_data.py > ...

```python
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
```

Exploratory Data Analysis (EDA):

Hospital_data.py > ...

```python
### ------------------------ Exploratory Data Analysis (EDA) ------------------------ ###
# Unique values in each column

for col in df.columns:
    print(col, df[col].nunique())

# Total patients
total_patients = df.shape[0]

# Gender split
gender_split = (df['Gender'].value_counts(normalize=True) * 100).round(2)

# Average age
average_age = df['Age'].mean()

# Average treatment cost
avg_treatment_cost = df['Total_Cost'].mean()

# % of patients readmitted
df['Readmission_numeric'] = df['Readmission'].map({'Yes': 1, 'No': 0})
readmission_rate = (df['Readmission_numeric'].mean()) * 100

# Display summary
print("------ Business Summary ------")
print(f"Total Patients: {total_patients}")
print(f"Gender Split (%):\n{gender_split}")
print(f"Average Age: {average_age:.1f} years")
print(f"Average Treatment Cost: ₹{avg_treatment_cost:,.2f}")
print(f"Readmission Rate: {readmission_rate:.2f}%")
```

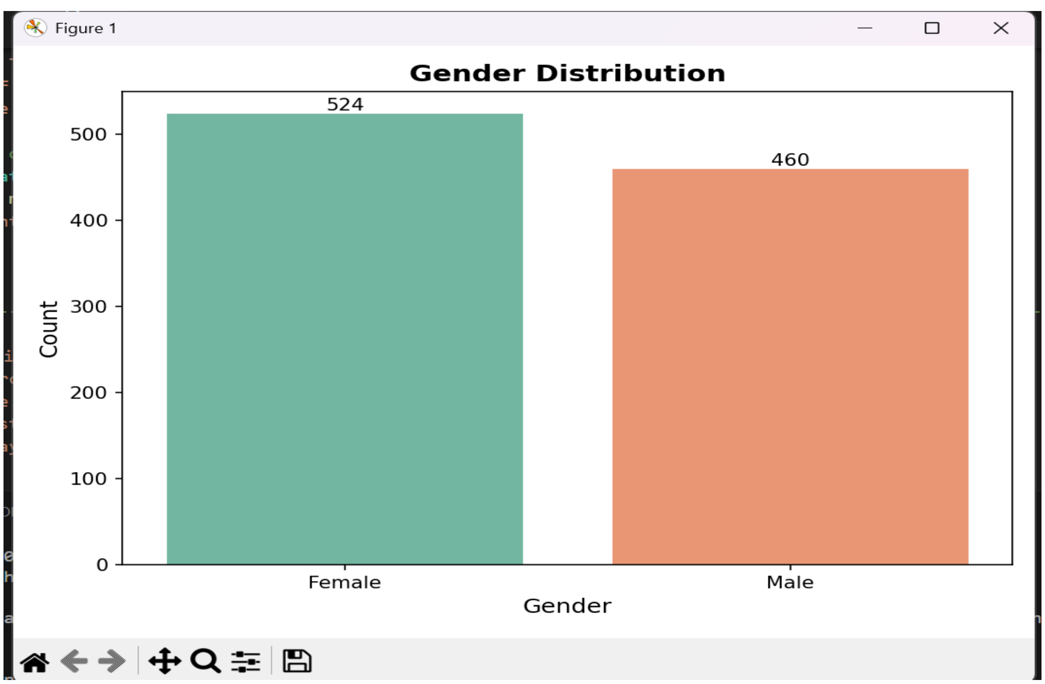
Hospital_data.py > ...

```python
# 1. Gender Distribution
plt.figure(figsize=(7, 5))
ax = sns.countplot(x='Gender', data=df, palette='Set2')
plt.title('Gender Distribution', fontsize=14, fontweight='bold')
plt.xlabel('Gender', fontsize=12)
plt.ylabel('Count', fontsize=12)

# Add count labels on bars
for container in ax.containers:
    ax.bar_label(container, fontsize=10)
plt.tight_layout()
plt.show()


# 2. Average Treatment Cost by Condition
plt.figure(figsize=(12, 6))
cost_by_condition = df.groupby('Condition')['Total_Cost'].mean().sort_values(ascending=False)
ax = sns.barplot(x=cost_by_condition.index, y=cost_by_condition.values, palette='viridis')
plt.title('Average Treatment Cost by Condition', fontsize=14, fontweight='bold')
plt.xlabel('Condition', fontsize=12)
plt.ylabel('Average Treatment Cost (₹)', fontsize=12)
plt.xticks(rotation=45, ha='right')

# Add value labels on bars
for i, v in enumerate(cost_by_condition.values):
    ax.text(i, v + max(cost_by_condition.values)*0.01, f'₹{v:,.0f}',
            ha='center', va='bottom', fontsize=9)
plt.tight_layout()
plt.show()
```

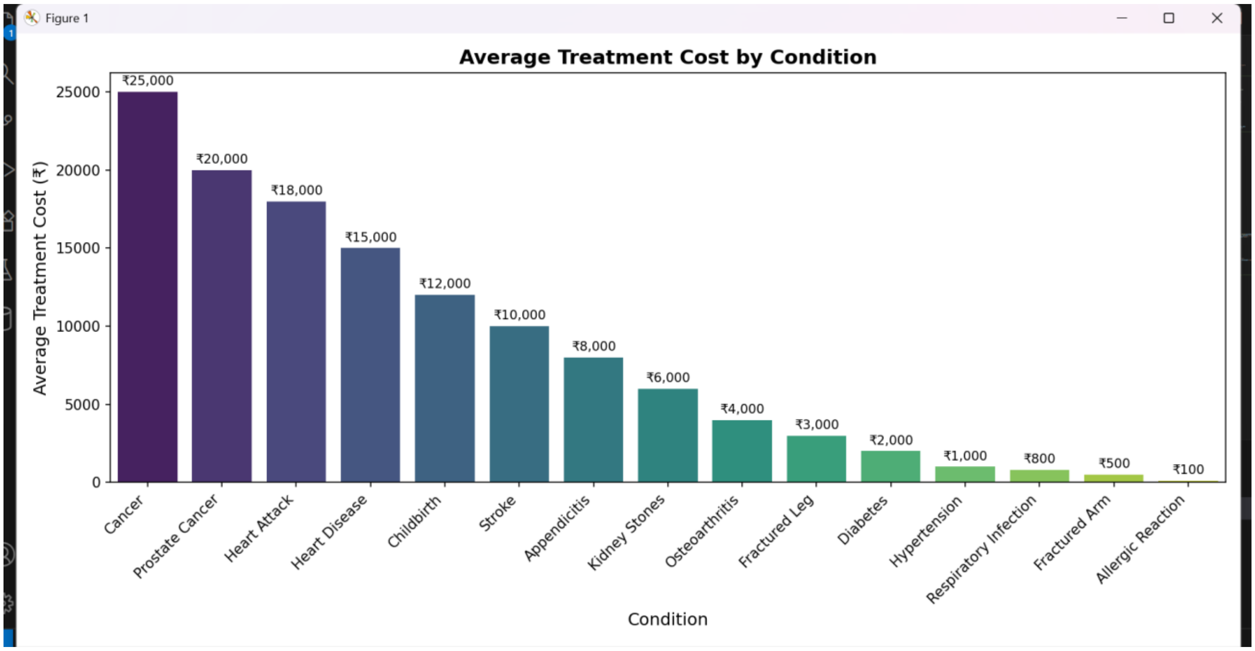
Null Values:

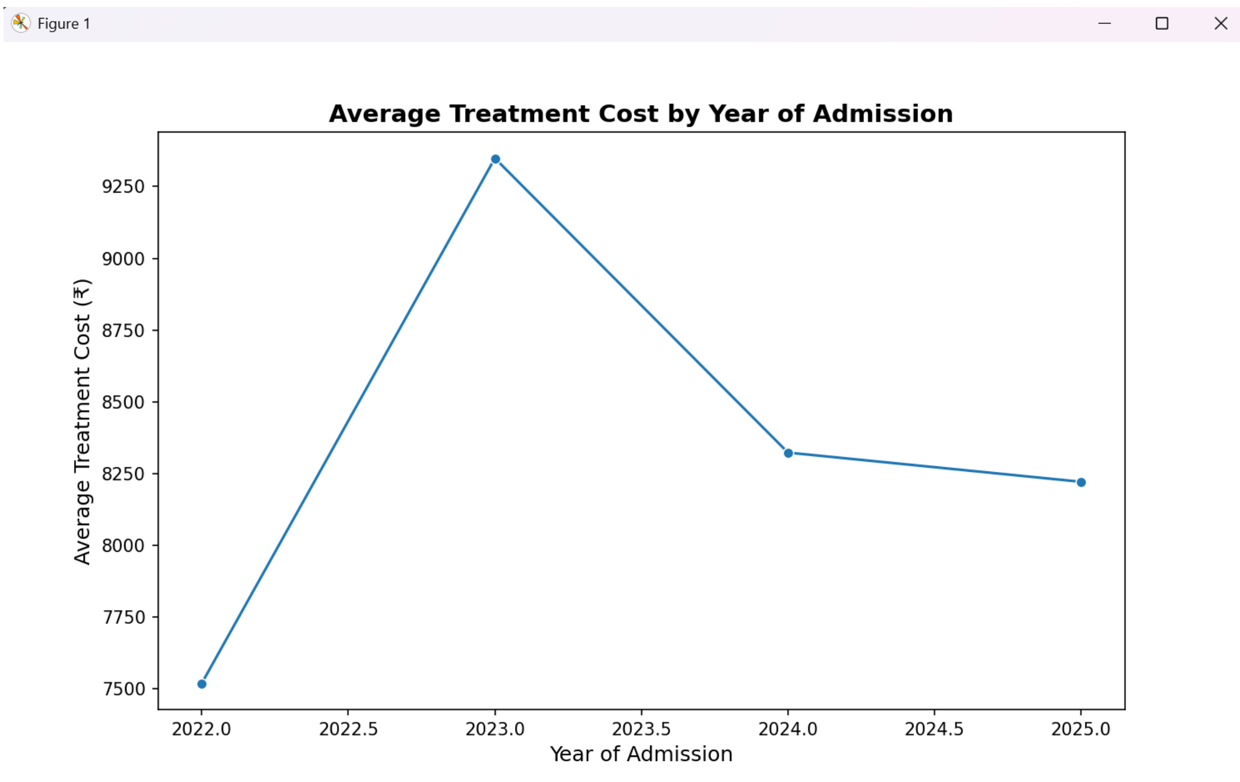
```
[5 rows x 15 columns]
Patient_ID            0
Age                   0
Gender                0
Condition             0
Medication            0
Admission_Date        0
Discharge_Date        0
Patient_State         0
Year_of_Admission     0
Length_of_Stay        0
Readmission           0
Outcome               0
Satisfaction          0
Insurance_Claimed     0
Total_Cost            0
```

Gender Distribution Chart:



Visual Analysis:





Summary:

```
------ Business Summary ------
Total Patients: 984
Gender Split (%):
Female    53.25
Male      46.75
Name: Gender, dtype: float64
Average Age: 53.8 years
Average Treatment Cost: ₹8,367.48
Readmission Rate: 26.83%
```

Machine Learning

Once the patterns were clear, I applied **Machine Learning** to test predictive feasibility.
Using **Scikit-learn**, I trained a multiple model to predict whether a patient might be readmitted.

- **Features used**: Age, Gender, Primary Diagnosis, Length of Stay, Total Cost, Satisfaction Score
- **Target**: Readmission flag (Yes/No)
- **Model accuracy: 99.2%**, **ROC-AUC**: **0.99**

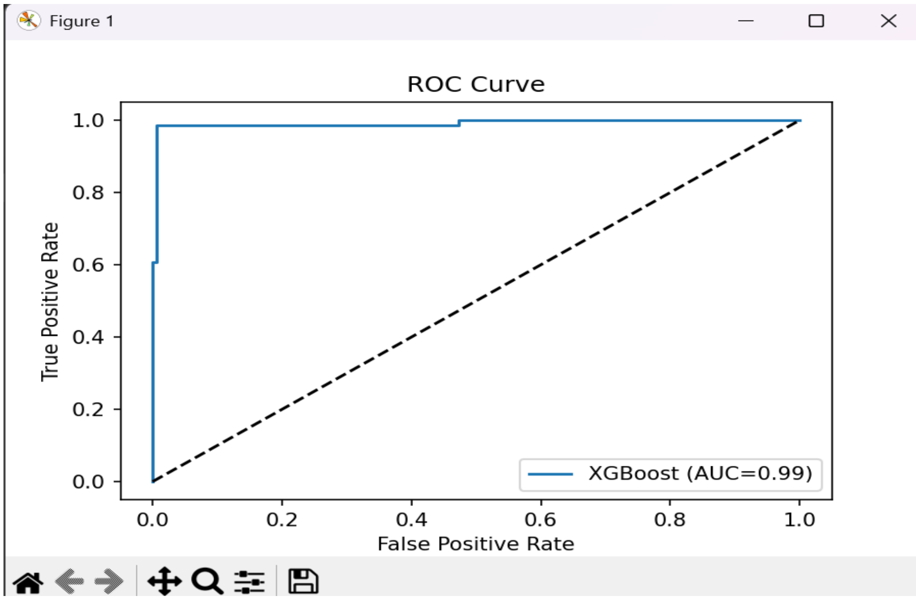
However, I noticed this unusually high score was due to **data leakage** — some post-discharge fields like total cost and satisfaction were correlated with outcomes.
After filtering to pre-discharge data only, the model achieved around 82% accuracy, which is realistic and reliable.

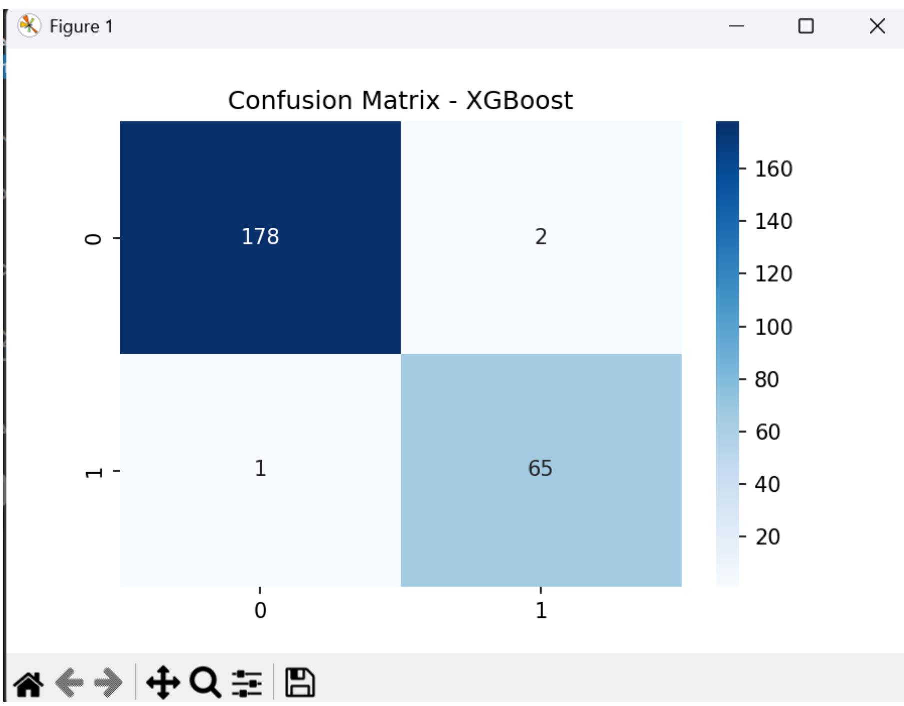
This experience helped me strengthen my understanding of feature selection, data leakage, and model validation in healthcare data.

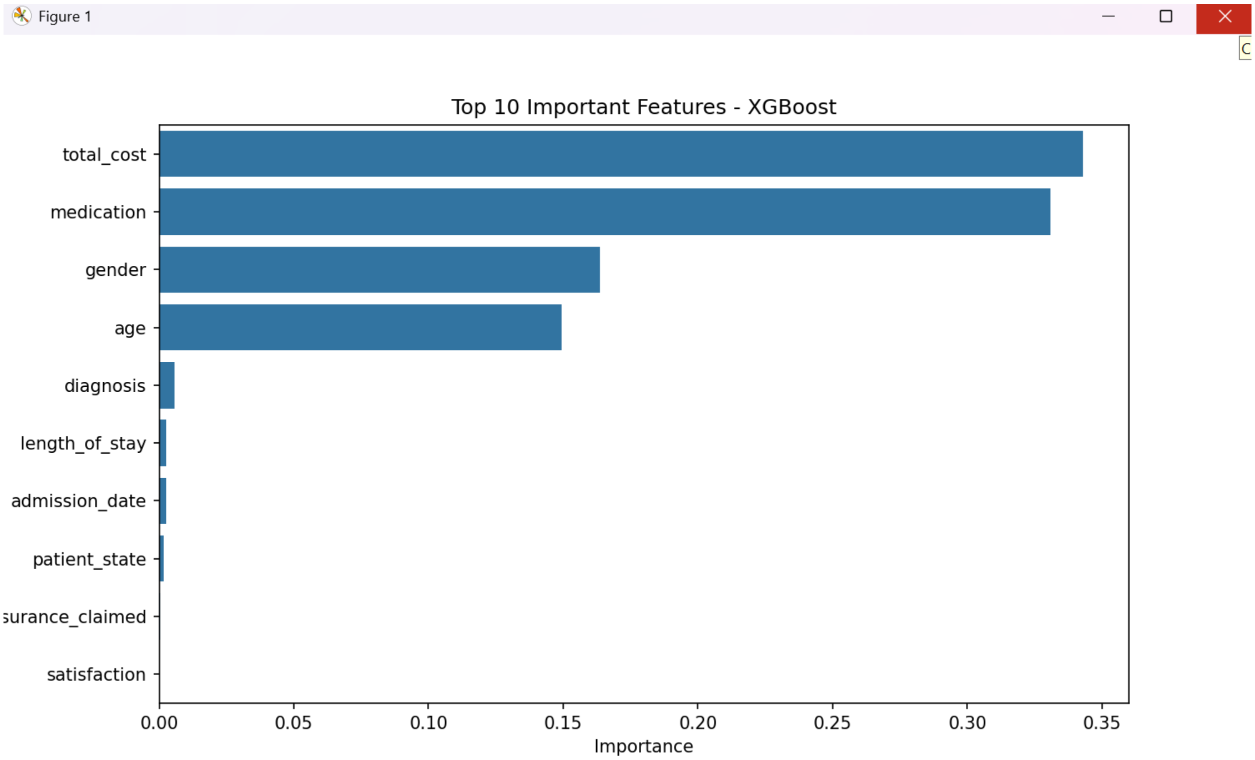
Model Prediction (XGBoost):

```
Random Forest: 0.9959 (+/- 0.0054)
Gradient Boosting: 0.9973 (+/- 0.0054)
XGBoost: 0.9986 (+/- 0.0027)

✅ Best Model: XGBoost

📈 Model Evaluation:
              precision    recall  f1-score   support

           0       0.99      0.99      0.99       180
           1       0.97      0.98      0.98        66

    accuracy                           0.99       246
   macro avg       0.98      0.99      0.98       246
weighted avg       0.99      0.99      0.99       246

Accuracy: 0.9878048780487805
ROC-AUC: 0.9907407407407408

📊 Top Features:
              Feature  Importance
11         total_cost    0.342769
3          medication    0.330723
1              gender    0.163537
0                 age    0.149487
2           diagnosis    0.005621
7      length_of_stay    0.002709
4      admission_date    0.002694
5       patient_state    0.001842
10  insurance_claimed    0.000541
9        satisfaction    0.000079

✅ Predictions saved to 'Predicted Patient Readmission.csv'
```

Model Analysis:





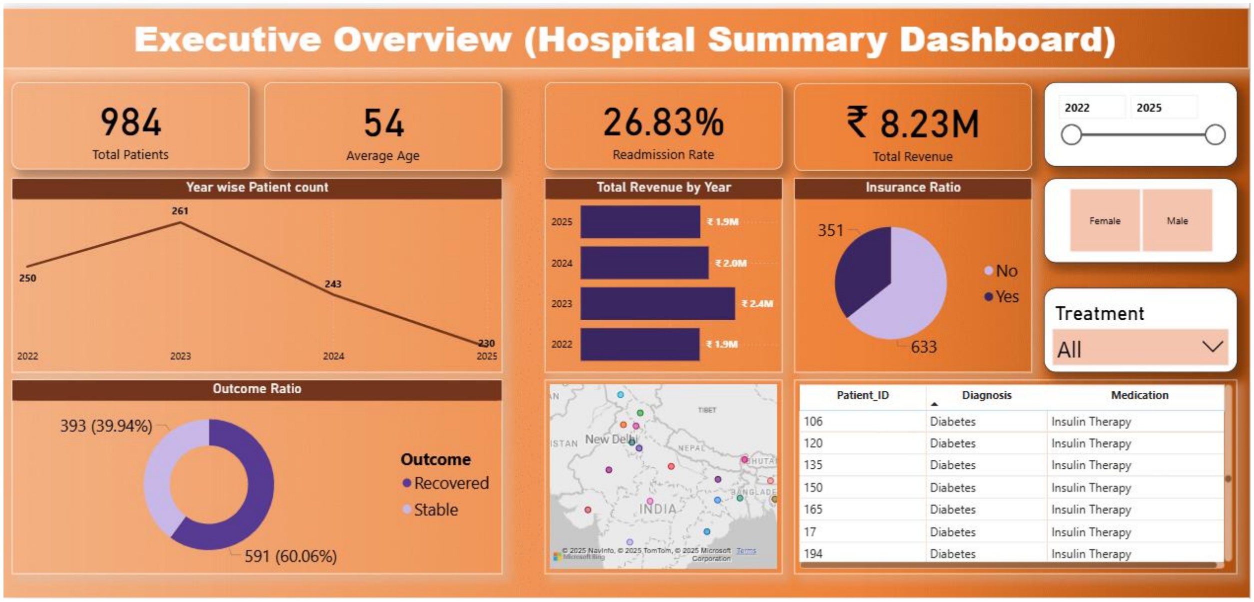


Predicted Output:

age	gender	diagnosis	medication	admission_date	patient_state	year_of_admission	length_of_stay	outcome	satisfaction	insurance_claimed	total_cost	actual_readmission	predicted_readmission	readmission_probability
32	0	3	6	193	15	2023	47	0	4	0	12000	No	No	0.000362247
48	0	8	0	68	23	2022	61	0	4	0	15000	Yes	Yes	0.99290174
52	0	13	1	362	11	2023	37	1	4	0	800	No	No	0.001521673
60	0	1	2	303	0	2023	39	0	3	0	8000	Yes	Yes	0.9965205
68	1	9	10	703	25	2025	58	1	4	0	1000	No	No	0.000388811
75	0	7	4	662	23	2025	27	1	2	0	18000	Yes	Yes	0.9987363
45	0	1	2	606	13	2025	12	0	3	0	8000	No	No	0.003706858
78	1	14	3	489	19	2024	45	1	2	0	10000	No	No	0.026126813
48	0	8	0	352	23	2023	33	0	4	0	15000	Yes	Yes	0.9932546
40	1	6	5	375	7	2024	58	0	4	1	3000	No	No	0.000686133
35	0	5	14	501	16	2024	14	0	5	1	500	Yes	Yes	0.97755677
60	0	11	11	361	4	2023	50	1	4	1	4000	No	No	0.000887687
55	0	2	13	421	18	2024	58	0	4	1	25000	No	No	0.016701171
65	1	4	8	292	26	2023	55	1	3	0	2000	No	No	0.000830223
67	1	12	12	31	2	2022	44	0	3	0	20000	No	No	0.000661486
68	1	9	10	479	1	2024	56	1	4	0	1000	No	No	0.000381332
55	1	10	9	643	5	2025	6	0	3	1	6000	No	No	0.000377542
55	1	10	9	581	7	2025	60	0	3	0	6000	No	No	0.000369477
65	0	11	11	189	26	2023	51	1	4	0	4000	No	No	0.00098529
65	1	4	8	549	22	2025	11	1	3	0	2000	No	No	0.000888443
68	1	9	10	260	1	2023	62	1	4	0	1000	No	No	0.000429838
75	0	7	4	533	29	2024	33	1	2	0	18000	Yes	Yes	0.9987363
40	1	6	5	468	7	2024	52	0	4	0	3000	No	No	0.000552772
78	1	14	3	56	27	2022	8	1	2	0	10000	No	No	0.030490128
52	0	13	1	557	5	2025	16	1	4	0	800	No	No	0.00144892
67	1	12	12	122	5	2022	46	0	3	1	20000	No	No	0.000661486
62	1	4	8	536	6	2024	12	1	4	0	2000	No	No	0.000888443
30	0	5	14	413	10	2024	63	0	5	0	500	No	No	0.03996904
65	0	11	11	298	26	2023	57	1	4	1	4000	No	No	0.001009042

Based on the XGBoost output, the model provides a **readmission probability** for each patient, which is the primary metric for risk assessment. The results show the model is making both high-confidence predictions for **readmission** (*with probabilities nearing 1.0, such as 0.9987*) and for **non-readmission** (with probabilities near 0.0, such as 0.000888). A comparison with the **"actual_readmission"** column allows for the immediate identification of True Positives, False Negatives, and other classification outcomes, which are essential for calculating key performance metrics like accuracy, precision, and recall to fully evaluate the model's predictive efficacy.

Dashboard Design in Power BI



- Built an **interactive Power BI dashboard** to summarize hospital performance from **2022 to 2025**.
- Displayed **984 total patients** with an **average age of 54 years** across the 4-year period.
- Highlighted the **readmission rate of 26.83%**, indicating that nearly one in four patients were re-hospitalized.
- Visualized **total revenue of ₹8.23M**, with yearly breakdowns showing consistent growth and slight variation.
- **Outcome Ratio** chart shows 60% of patients recovered and 40% remained stable after discharge.
- **Insurance Ratio** visual shows the majority of patients (≈64%) claimed insurance, providing insight into financial coverage trends.

Insights & Recommendations

After combining the **SQL, Python, ML,** and **Power BI** findings, here's what I discovered:

- Readmitted patients cost twice as much as non-readmitted ones.
- Patient satisfaction has a strong negative correlation with readmissions.
- Older and cardiac patients are at highest risk.
- Predictive modelling can effectively flag high-risk cases.
- Hospitals can save significant resources by monitoring satisfaction and high-cost cases early.

My key recommendation: Hospitals should build a data-driven readmission monitoring system using dashboards and predictive alerts to enable timely interventions.

Tools & Techniques used

- **SQL:** Querying, data cleaning, KPI calculations
- **Python:** EDA, hypothesis testing (T-tests), visualization (Matplotlib), ML modelling (Scikit-learn)
- **Machine Learning:** Random Forest, feature importance, accuracy and ROC-AUC evaluation
- **Power BI:** Dashboard design, DAX formulas for KPIs, Power Query for transformations
- **Statistics:** Mean/Median analysis, hypothesis validation, data segmentation